

Robert Mostoghiu Paun

Data scientist | Computational Astrophysicist

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Summary

Results-oriented Data Scientist with 4+ years of experience and a PhD in Computational Astrophysics

Expert in designing, orchestrating, and validating large-scale predictive models and automated ETL pipelines for terabyte-scale datasets.

Proven track record of translating complex technical architectures into actionable business insights and high-performance software solutions.

Skills

Programming & Databases: Python (NumPy, SciPy, Pandas/Polars, Seaborn, Matplotlib), Jupyter, SQL, R, C/C++.

Data Science & Modelling: Numerical & Predictive Modelling, Machine Learning (scikit-learn, PyTorch), AI Orchestration, Large-Scale Data Analysis, Data Visualisation.

Tools & Technologies: High-Performance Computing (HPC), Cloud Services (Azure Databricks), Dashboards (Power BI, Excel), Git, Linux, LaTeX.

Professional: Project Management, Stakeholder Communication, Cross-Functional Leadership, Research & Problem Solving.

Professional History

Postdoctoral Research Associate at Swinburne University

Jan 2022 – Present

- **Model optimisation:** Led an international project to develop a novel simulation technique, **improving model accuracy by 40%** and correcting critical computational errors in large-scale predictive models.
- **End-to-End Project Management:** Managed the full lifecycle of complex data projects, from conceptual design and HPC execution to final analysis and delivery.
- **Data Engineering & ETL:** Designed and implemented automated ETL pipelines using Python and SQL to transform terabyte-scale raw simulation data into structured, analysis-ready formats.
- **Data Quality & Integrity:** Developed data science techniques to identify anomalies and distinguish signal from computational noise, ensuring high-fidelity data for decision-making.

- **Stakeholder Engagement:** Delivered weekly performance reports and data visualisations to internal stakeholders, ensuring alignment on complex technical milestones.
- **Leadership & Mentoring:** Supervised undergraduate and postgraduate researchers and assumed leadership roles within the research centre to foster collaborative environments.

Projects

GIZMO Simulation Code: Custom Physics Model

Tech Stack: C, Python, Fortran, Git

- Implemented a novel physical model into a complex cosmological codebase, extending its predictive capabilities.
- Optimised a legacy C codebase for parallelised HPC frameworks, improving simulation efficiency and stability.
- Managed the full development lifecycle via Git, demonstrating the ability to navigate and extend large-scale legacy systems.

SAGE AI-First Merger Tree Converter

Tech Stack: Python, LLM CLIs, Git

- Engineered an AI-driven data orchestration tool using Python and LLM CLIs to automate the standardisation of complex hierarchical datasets for integration with predictive models.
- Developed automated ETL pipelines that leverage AI-driven logic to handle diverse data schemas, optimising I/O and memory management for increased processing efficiency at scale.

Education

PhD in Computational Astrophysics from Universidad Autónoma de Madrid

November 2020

Awards: Summa Cum Laude (2020), Doctorate Honourable Mention (2022).

Professional Development

Research Translation & Design Thinking from IdeaSquare (CERN)

July 2025

Commercialisation for Medical Researchers from AAMRI VIC

September – December 2025

Languages

English, Spanish, Romanian